CS & IT ENGINEERING

Algorithms

Analysis of Algorithms

Lecture No.- 03

By- Aditya sir

Recap of Previous Lecture

Topic

Topic

Apriori Analysis

→ B.C

→ W.C

→ A.C

Topics to be Covered

Topic

Topic

Topic

Asymptotic Notations

About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.

Telegram

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Topic : Asymptotic Notations

Time complexity (n)

logarithmic	Polynomial	Exponential
	n^x	a^n
$\log(n)$	$n, n^2, \sqrt{n}, n^3$	$2^n, 3^n \ldots n^n$
$\log(\log n)$		

Topic : Asymptotic Notations

Asymptotic Notations:

- The bounds/range of the function can be represented using these notations.

Topic : Asymptotic Notations

Types of Bounds:

max

(1) Upper Bound

(2) Lower Bound

min

(i) Lose Bound

(ii) Tight Bound

min max

	Loose	Tight
U	✓	✓
L	✓	✓

Topic : Asymptotic Notations

Types of Bounds:

>

Loose LB

Loose UB

1 . . . . . 66 67 68 69 ← $\boxed{70}$ → 71 72 73 100

Lower Bound (LB)

Upper Bound (LB)

Tight LB

Tight UB

Tight → closest

Topic : Asymptotic Notations

V.V. Imp

Asymptotic Notations

(1) Big Notations
- (i) UB : Big Oh → O
- (ii) LB : Big Omega → Ω

Loose + Tight

(2) Small Notations — Little
- (iii) UB : Small oh → o
- (iv) LB : Small omega → ω

loose

(v) Theta : θ → Tight Bound

Closest

Topic : Asymptotic Notations

Let 'f' and 'g' be functions from the set of integers/real to real number;

1. **Big-Oh(O):** Upper Bound (UB)

- f(n) is O(g(n)) if there exists $\boxed{\text{some}}$ constant $c > 0$ and $n_0 > 0$ such that $f(n) \le c \cdot g(n)$ ~~c.g(n)~~, whenever $n \ge n_0$.

$$f(n) = O(g(n))$$

iff $\underline{f(n) \le c * g(n)}$

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Example:

(1) Order of Magnitude

$$\boxed{f(n) = n^2 + n + 1}$$

$$\cancel{1 + n \leq n^2 + n^2}$$

$$\cancel{n = 2 \quad 1 + 2 \leq 2^2 + 2^2}$$

$$\cancel{n = 3 \quad 1 + 3 \leq 3^2 + 3^2}$$

$$1 \leq n^2$$

$$1 + n \leq n^2 + n^2$$

$$1 + n + n^2 \leq n^2 + n^2 + n^2$$

$$\boxed{1 + n + n^2 \leq 3n^2}$$

$$f(n) \leq c * g(n)$$

Hence,

$$\boxed{f(n) = O(g(n))}$$

$$\underline{1 + n + n^2 = O(n^2)}$$

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- Whenever we determine the upper bound and lower bound, we should find that function 'g' which is closest to the given function.

Example:

$$\text{Nagpur} \xrightarrow[\text{(Non-stop)}]{\text{Flight}} \text{Delhi}$$

$$\left.\begin{array}{l} \text{Loose UB} \begin{bmatrix} P1 \rightarrow < 1 \text{ year} \\ P2 \rightarrow < 1 \text{ week} \end{bmatrix} \\ \boxed{\text{Tight}} \text{ UB} \begin{bmatrix} P3 \rightarrow < 5 \text{ hr} \end{bmatrix} \end{array}\right] \text{UB}$$

Topic : Asymptotic Notations

Example2:

Friends

Loose LB

F1 : > 50 Rs
F2 : > 5000 Rs

LB

Tight LB

F3 : > 5L Rs

Topic : Asymptotic Notations

Shortcut:

Dominating term → ~~Highest~~ term with highest rate of growth

Example:

(i) $f(n) = n^2 + n + 1$

$f(n) = O(n^2)$

$1 + n + n^2 \le 10n^2$

(ii) $f(n) = \cancel{5}n^3 + \cancel{8n + 7}$

$f(n) = O(5n^3) = O(n^3)$

why?

2025 ⟶ 50K

2027 ⟶ 2L

2030 ⟶ 50L

2035 ⟶ $\boxed{100cr}$ ✓

~ 100cr

Topic : Asymptotic Notations

(1) Step-count method

$T(n) = \underline{4n^2 + 8n + 8}$

$T(n) = O(4n^2)$

$\boxed{T(n) = O(n^2)} \rightarrow$ Tight UB

(2) Order of magnitude

$T(n) = n^2 + n + 1$

$\boxed{T(n) = O(n^2)} \rightarrow$ Tight

$n^2 + n + 1 \leq 10n^3 \rightarrow O(n^3)$

$\rightarrow O(n^4)$

$\rightarrow O(n^5)$

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2. Big-Omega(Ω): Lower Bound (LB)

- f(n) in $\Omega(g(n))$ iff there exists some constant 'c' and 'n_0' such that $f(n) \geq c.g(n)$, whenever $n \geq n_0$.

$f(n) = \Omega(g(n))$

iff, $f(n) \geq c.g(n)$; $n \geq n_0$

Topic : Asymptotic Notations

Example: $8n^2 + 3n + 5$

$f(n) = 1 + n + n^2 \geq 1 * n^2 \rightarrow \Omega(n^2)$

$f(n) = 1 + n + n^2 \geq 1 * n \rightarrow \Omega(n)$

$f(n) = n + n + n^2 \geq 1 * \sqrt{n} \rightarrow \Omega(\sqrt{n})$

$f(n) = n + n + n^2 \geq 1 * 1 \rightarrow \Omega(1)$

By default → Tight Bound ✓

$1 + n + n^2 \geq 1 * n^2$

$f(n) \geq c.g(n)$

$f(n) = \Omega(g(n)) = \Omega(n^2)$

Hence,

$$\boxed{1 + n + n^2 = \Omega(n^2)}$$

Topic : Asymptotic Notations

3. **Theta(θ):** Tight Bound

- $f(n)$ is $\theta(g(n))$ iff $f(n)$ is $O(g(n))$ and $f(n)$ is $\Omega(g(n))$

  $c_1 . g(n) \le f(n) \le c_2 . g(n)$

If $f(n) = O(g(n))$

and

$f(n) = \Omega(g(n))$

$\Rightarrow f(n) = \theta(g(n))$

Topic : Asymptotic Notations

Example: $n^2 + n + 1$

$$1 * n^2 \leq \underbrace{1 + n + n^2}_{f(n)} \leq 3 * n^2$$

$$c_1 . g(n) \leq f(n) \leq c_2 . g(n)$$

$f(n) = O(n^2)$

and

$f(n) = \Omega(n^2)$

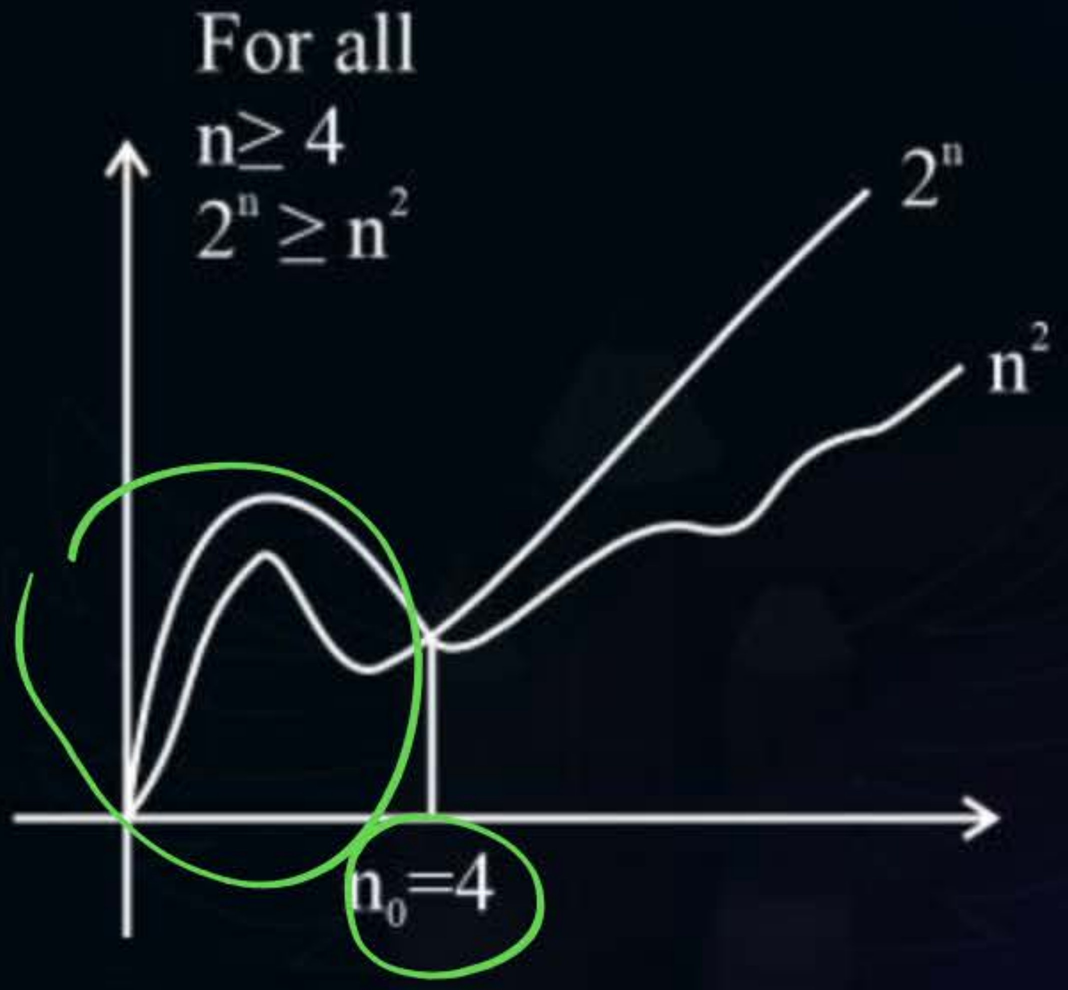
$\rightarrow$ $\boxed{f(n) = \theta(n^2)}$ $\rightarrow$ Tight bound

Topic : Asymptotic Notations

Example: n^2 vs 2^n

$n^2 \le c * 2^n$ $\quad$ $n^2 = O(2^n)$ ✓

n	n^2		2^n
1	1	<	2
2	4	=	4
3	9	>	8
4	16		16
5	25		32
6	36		64
7	49		128



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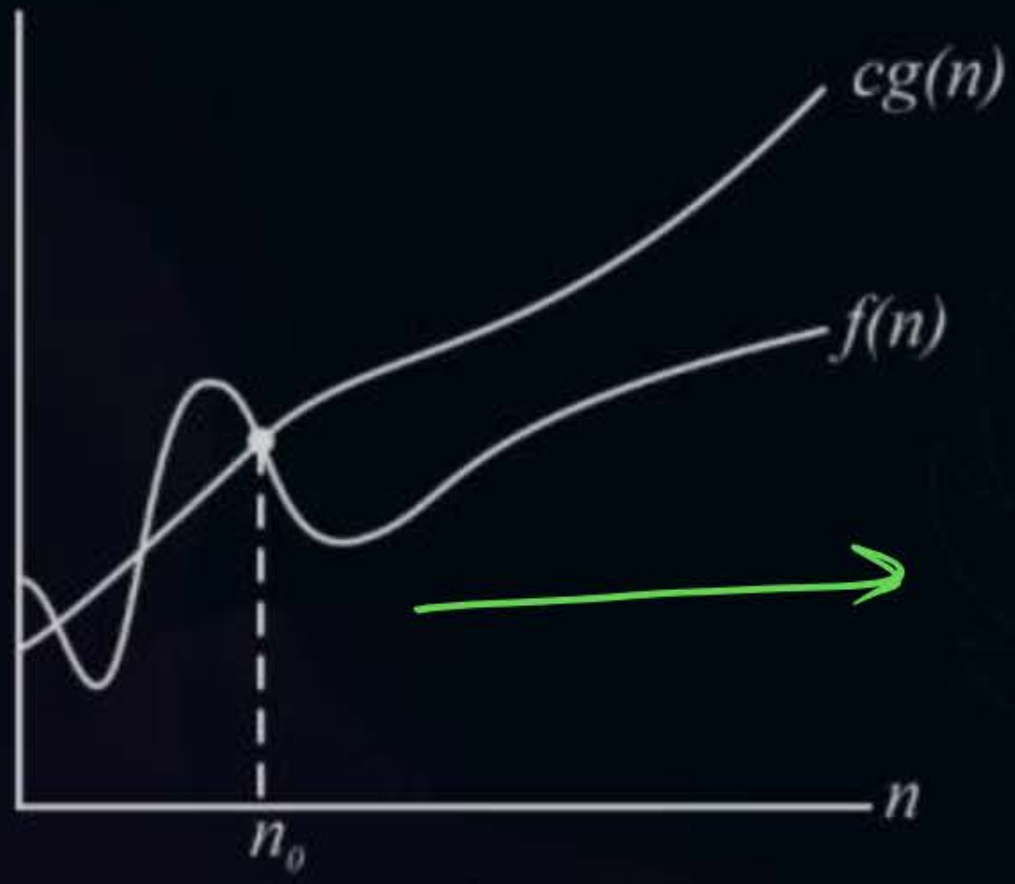
Big-Oh Notation:

$$f(n) \le c * g(n);\ n \ge n_0$$

$$f(n) = O(g(n))$$

$n \ge n_0$



$f(n) = O(g(n))$

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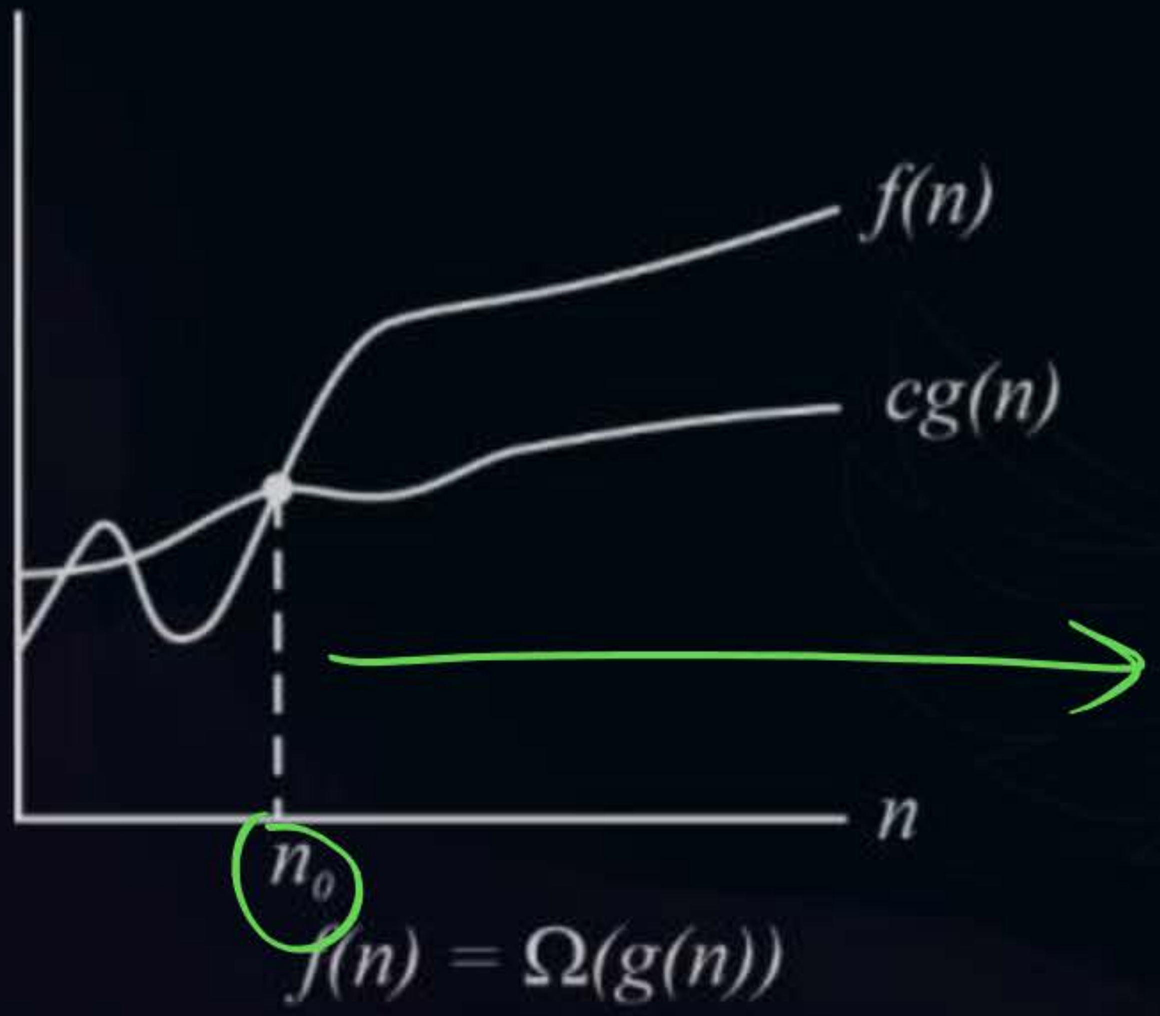
Big-Omega Notation:

$$f(n) \geq c * g(n);\ n \geq n_0$$

$$f(n) = \Omega(g(n))$$



$f(n) \geq c * g(n)$

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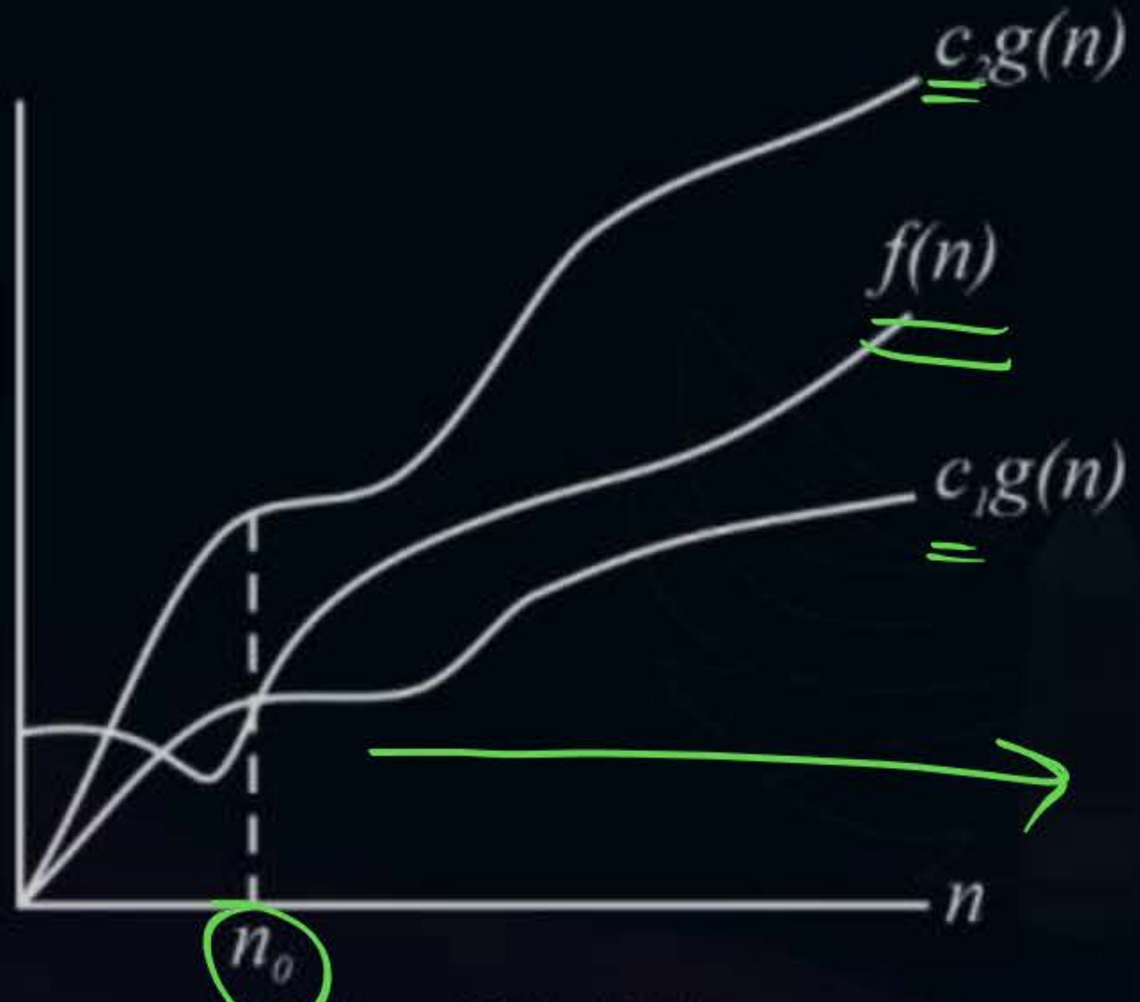
Theta Notation:

$$c_1 * g(n) \le f(n) \le c_2 * g(n);\ n \ge n_0$$

$$f(n) = \theta(g(n))$$



$f(n) = \Theta(g(n))$

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Summary:

Asymptotic Notations

Big

O, Ω θ

Small

o, ω

larger = $\underline{\ \Omega\ }$ (smaller)

$n^2 \geq 1 * n$

$n^2 = \Omega(n)$

$2^n = \Omega(n)$

$2^n = \Omega(n^2)$

$$Small = \underline{O}(large)$$

$$n = O(2^n)$$

$$\underline{n^2 = O(2^n)}$$

Topic : Asymptotic Notations

Important observations:

(1) ~~Longer~~ Larger functions are always omega of the smaller functions.

$2^n = \Omega(n^2)$

$2^n = \Omega(n^3)$

e.g.: $n^3 \geq c * n^2$

$n^3 = \Omega(n^2)$

Topic : Asymptotic Notations

Important observations:

(2) Smaller functions are always order (Big Oh) of the larger functions.

$n^2 = O(2^n)$

$n^3 = O(2^n)$

e.g.: $n^3 \geq c * n^2$

$n^2 = O(n^3)$

Important observations:

(3) If two functions have equal rate of growth then they are theta of each other.

$f(n) = 8n^2$

$g(n) = 5n^2$

e.g.: $\boxed{\begin{array}{c} f(n) = \theta(g(n)) \\ \text{or} \\ g(n) = \theta(f(n)) \end{array}}$

V. Imp: Asymptotic vs Mathematical

$f = 8n^2$

$g = 5n^2$

$\boxed{f = \Theta(g)}$

$f =_A g$

$8n^2 = 5n^2$?

$\hookrightarrow$ No

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Practice Questions:

(1) $f(n) = 8n$

$$f(n) = 8n$$

$$8n \leq 10n \Rightarrow O(n)$$

$$8n \geq 2n \Rightarrow \Omega(n)$$

$$\underline{\underline{\Theta(n)}}$$

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Practice Questions:

(2) $f(n) = 9^{200}$

9^{200} ⟶ Constant (independent of n)

→ $O(1)$

→ $\Omega(1)$

$O(1)$, $\Omega(1)$ → $\Theta(1)$

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Practice Questions:

(3) $f(n) = \underline{\underline{100n^2}} + \underline{\underline{500\ n}}$

$\downarrow$

dominating term

$$f(n) = O(n^2)$$
$$= \Omega(n^2)$$
$$\Theta(n^2)$$

Topic : Asymptotic Notations

Practice Questions:

(4) $f(n) = 100(\log n) + 50 * \sqrt{n}$ $\longrightarrow$ $\Theta(\sqrt{n})$

$O(\sqrt{n})$ $\quad$ $\Omega(\sqrt{n})$

$\boxed{\log n \quad \text{vs} \quad \sqrt{n}}$

Appr 1 :- Substitution mtd

$n = 64$

$\log_2 n \quad \underset{\text{vs}}{<} \quad \sqrt{n}$ → dominating

$\log_2 (2^6) \quad \sqrt{64}$

$6 < 8$

Approz :- Taking log( ) both sides

(only when unable to solve directly)

$$\log n < \sqrt{n} = n^{1/2}$$

$$\log(\log n) \qquad \log(n^{1/2})$$

$$\log(\log n) < \frac{1}{2}\log n$$

eg:- $n^2 < n^3$

$\log(n^2) \quad \log(n^3)$

$2\cancel{\log}n \quad 3\cancel{\log}n$

$2 < 3$

$O(1) \quad \times \quad O(1)$

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Practice Questions:

(5) $f(n) = 500 * \sqrt{n} + 2n + 100$

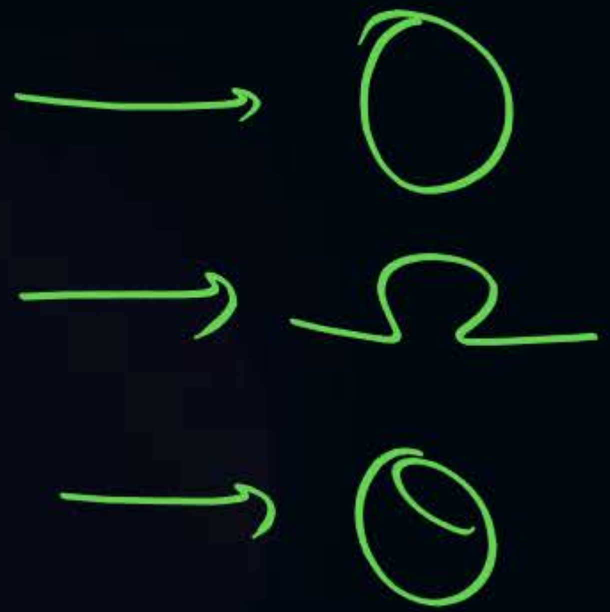
$= O(n)$

$= \Omega(n)$ $\Rightarrow \Theta(n)$

$\sqrt{n}$ vs n

$\sqrt{n} < n$

Summary



THANK - YOU